
Compete or Coevolve: An Evolutionary Macro-Micro Framework for AI-Era Educational Policy

Summary

As the proverb goes, ‘When the winds of change blow, some build walls, while others build windmills.’ In an era where the rapid iteration of Generative AI outpaces traditional academic cycles, the widening gap between static curricula and dynamic labor demands poses an existential challenge to higher education. Our work investigates the systemic impacts of Gen-AI on occupations and post-secondary education, aiming to support adaptive and sustainable institutional strategies.

For Task 1, we select three representative careers—Data Scientist (STEM), Electrician (Trade), and Graphic Designer (Arts)—and construct a **task-based exposure framework** to characterize how Gen-AI affects labor demand. Using ONET task data and text embedding techniques, we assess AI’s impact through substitution, augmentation, and immunity. Forecasts reveal divergent outcomes: a **widening talent gap** in STEM fields, **dynamic balance** for electricians, and **structural unemployment** among graphic designers.

For Task 2, we develop the **Evolutionary Macro–Micro Coupled Adaptive System** (E-MMCAS) to simulate dynamic educational policy interventions. At the macro level, a modified **Lotka–Volterra** system models AI–labor interactions, while at the micro level, curricula are represented as course–skill bipartite networks. Curriculum reform is modeled as network rewiring, and a bi-level Discrete Multi-objective Particle Swarm Optimization (DMPSO) algorithm **provides tailored policy recommendations for three types of institutions**.

For Task 3, we construct a **Holistic Evaluation Module integrating Monte Carlo simulation**, entropy-based weighting, and the **VIKOR** compromise decision method. **Six indicators** (employability, adaptability, innovation, interdisciplinarity, environmental impact, and ethical compliance) are evaluated under uncertainty. The results identify AI-orchestrated STEM strategies as the best compromise solution, while vocational and arts pathways require targeted structural transformation to remain resilient.

For Task 4, we map optimized policies to interpretable institutional strategies for research universities, technical–vocational institutions, and arts schools. We show that resilient institutions shift **from task execution to system orchestration**, **reinforce physical skill immunity**, or redefine creativity through human judgment and ethical authorship. A **policy generalization principle** is proposed to enable transferability across educational archetypes.

In conclusion, generative AI does not make education irrelevant but changes its structure. Long-term career resilience depends not just on learning AI tools, but on redesigning skills to work alongside AI while keeping human judgment, creativity, and responsibility. Our framework offers a data-driven approach to guide education policy in the AI era.

Keywords: Tug-of-War Mechanism; DMPSO; Entropy Weighting Method; Macro-Micro Coupled Systems; Multi-Objective Optimization; VIKOR

Contents

1 Introduction	4
1.1 Problem Background	4
1.2 Restatement of the Problem	4
1.3 Our Work	5
2 Assumptions and Justifications	5
3 Notations	6
4 Data Collection and Preprocessing	6
4.1 Data Description	6
4.2 Textual Preprocessing	7
4.3 Numerical Preprocessing	7
5 Spatiotemporal Coupled Dynamics Model	7
5.1 Conceptual Framework: Decomposing the Impact of Gen-AI	7
5.2 Data Quantification	7
5.3 Model Formulation: The Tug-of-War Mechanism	8
5.3.1 The Net Impact Function $\Phi_i(t)$	8
5.3.2 The Coupled Dynamics	9
5.4 Simulation Results and Analysis	9
5.4.1 Trajectory Forecasting: A K-Shaped Future	9
5.4.2 Stability and Uncertainty Analysis	10
5.4.3 Conclusion	10
6 The E-MMCAS Framework for Strategic Employability Optimization ..	11
6.1 Macro-Dynamics: Lotka-Volterra Interaction Mechanism	11
6.2 Micro–Macro Coupling via Curriculum Network Rewiring	11
6.3 LogicStrategic Optimization Objective: Employability-First Design	13
6.4 DesignSolution Algorithm: Bi-Level Coupled DMPSO	13
6.5 The Simulation Results and Strategic Implications	14
6.6 Model Validation via Ablation Analysis	16
6.7 Conclusion	16
7 Multi-Objective Policy Evaluation under Uncertainty	16
7.1 Reframing the Problem: Policy Design under Conflicting Objectives	16
7.2 Calculation Transparency: Opening the Algorithmic Black Box	16
8 Empirical Results and Policy Selection	17
8.1 Simulation Design	17

8.2 Decision Matrix Construction	18
8.3 Entropy-Based Weighting Results	19
8.4 VIKOR Compromise Ranking	20
8.5 Interpretation of the Optimal Policy	20
9 Simulation Results and Institutional Strategies	20
9.1 STEM Institutions (MIT): From Code Writing to System Orchestration	20
9.2 Technical–Vocational Institutions (Lincoln Tech): Reinforcing the Physical Stronghold	21
9.3 Arts Institutions (RISD): From Artifact Creation to Meaning Orchestration	21
9.4 Synthesis: The Dissolution of Skill Barriers and the Return of Human Premium	21
10 Policy Generalization	22
10.1 Generalization Principle	22
10.2 Policy Transfer Across Institutional Archetypes	22
11 Sensitivity Analysis	22
12 Model Strengths and Weaknesses	23
12.1 Strengths	23
12.2 Weaknesses	23
13 Conclusion	24
References	25
Report on Use of AI	26
Utilized AI Techniques	26

1 Introduction

1.1 Problem Background

In recent years, Generative Artificial Intelligence (Gen-AI) has experienced rapid development, evolving from a niche experimental technology into a widely adopted tool embedded in everyday life. Advances in large language models, image generation systems, and automated content creation have significantly lowered the cost of performing cognitive tasks that once relied heavily on human labor. As a result, Gen-AI is increasingly influencing how work is performed across a wide range of professions, reshaping skill requirements, productivity levels, and employment structures [1].

Post-secondary educational institutions play a critical role in preparing future graduates for this evolving labor market shaped by Gen-AI [2]. Universities, trade schools, and arts institutions serve distinct populations and career pathways, and therefore face different challenges and opportunities in responding to the integration of this technology. Decisions regarding enrollment scale, curriculum design, and the incorporation of AI-related competencies will directly influence graduates' employability and long-term career resilience. Nevertheless, such decisions are often made in the absence of a systematic, data-informed understanding of how Gen-AI may alter specific professions over time.

Against this backdrop, a structured modeling approach is necessary to assess the potential impacts of Gen-AI on representative careers. Such an approach enables a systematic analysis of changes in professional demand, task composition, and skill requirements over time [3]. Based on these results, post-secondary institutions can develop actionable recommendations that better align academic programs with future workforce needs while balancing technological efficiency, human contribution, and broader societal considerations.

1.2 Restatement of the Problem

Considering the background, we need to solve the following problems:

- **Task 1** : Select three representative careers from STEM, trade, and arts fields, and construct a data-informed framework to characterize their task structures, core skill requirements, and career pathways, while modeling how Generative AI is expected to reshape task composition and labor demand over time.
- **Task 2** : Develop a dynamic policy simulation model to evaluate the long-term impacts of alternative AI education strategies under uncertainty in technological progress and institutional response.
- **Task 3** : Perform multi-objective policy evaluation and robustness optimization to identify compromise-optimal strategies balancing employability, adaptability, innovation, environmental sustainability, and ethical compliance.
- **Task 4** : Map the optimized policy outcomes to traditional institutional strategies, translating model results into interpretable and actionable recommendations for different types of post-secondary institutions.

1.3 Our Work

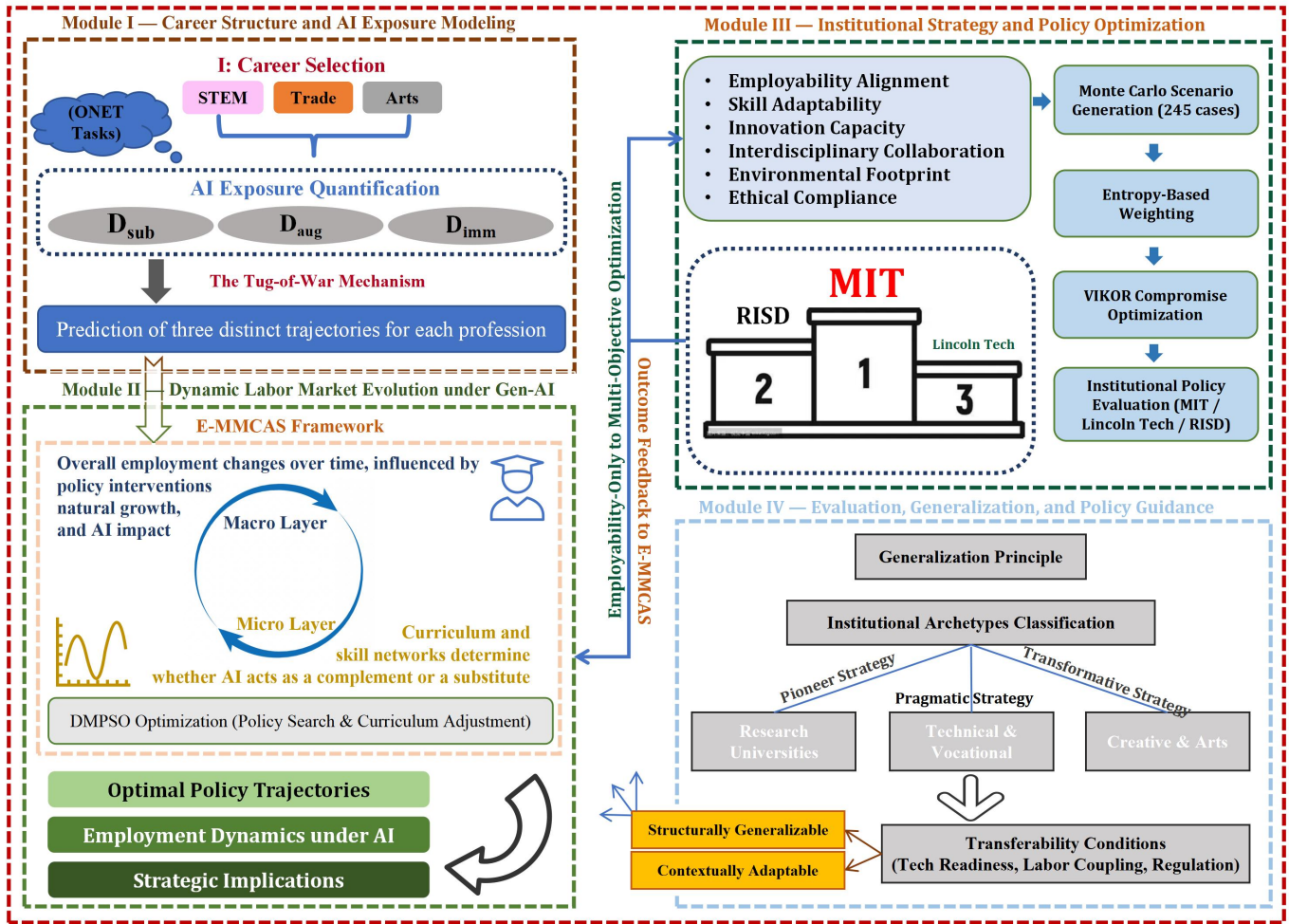


Figure 1. Our Work

2 Assumptions and Justifications

To simplify the problem, we make the following assumptions, each of which has a reasonable justification. Additional assumptions may be introduced later to facilitate analysis in specific sections.

- **Assumption 1:** The datasets used in this study are assumed to be sufficiently complete and representative of the selected careers and educational contexts. Data are collected from the U.S. Bureau of Labor Statistics, the ONET occupational information system, and official university websites, covering recent multi-year labor trends, task composition, and evolving skill requirements. Since the competition problem does not restrict the region, and U.S. labor data along with ONET 28.0 provide the most detailed publicly available classification of occupations, we use them as the baseline; the framework can later be adapted to other regions through parameter calibration.
- **Assumption 2:** We assume that generative AI primarily affects occupations at the task level, and only highly homogeneous, low-skill niche occupations may face full substitution.

The model focuses on representative occupations with employment sizes of 100,000 or more, excluding smaller-scale professions from the analysis. Tasks that are repetitive, information-based, or rule-driven are more likely to be automated or augmented, whereas tasks requiring physical presence, complex human interaction, or high-level creativity remain largely human-driven. Future AI capabilities are projected based on historical trends in hardware computing power and algorithmic advancement.

• **Assumption 3:** Post-secondary educational institutions are assumed to maintain stable institutional structures over the modeling horizon. While curricula and instructional content may adjust in response to Gen-AI, the overall program structure, training duration, and certification requirements are assumed to remain unchanged.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1. Notations used in this paper

Symbol	Description	Unit/Range
t	Time	Year
$D_i(t)/N_i(t)$	Labor Demand & Supply of occupation i at time t	Count
$D_{sub}/D_{aug}/D_{imm}$	Substitution/Augmentation/Immunity Drivers	[0,1]
$A(t)$	Generative AI Capability Index	[1,∞]
$\Phi_i(t)$	Net AI Impact Coefficient on labor market	[-1,1]
r_d	Natural market growth rate	1/Year
K	Carrying capacity of the labor market	Jobs
δ	Labor attrition rate (retirement / career shift)	1/Year
$u(t)$	Policy intervention intensity	Index
σ	Market volatility parameter	Index

4 Data Collection and Preprocessing

4.1 Data Description

The data for this study are drawn from three main sources. First, the ONET Database (version 28.0) serves as the primary benchmark, providing detailed task descriptions and occupational indicators such as abilities and work activities, which are used to model both physical and cognitive task requirements. Second, we collect curriculum information from representative post-secondary institutions, including MIT, Lincoln Tech, and RISD, to capture discipline-specific educational structures and course-level content. Third, external constraints are incorporated from diverse sources, including generative AI capability reports, energy consumption statistics, and legal-ethical precedent cases, to account for technological, environmental, and regulatory boundaries.

4.2 Textual Preprocessing

Traditional keyword-based methods are inadequate for capturing deep semantic relationships between tasks and AI capabilities. To address this, we implement a semantic quantification pipeline based on Sentence-BERT, mapping human task descriptions and AI capability descriptions into dense embedding vectors and measuring their similarity to estimate the likelihood that a task can be substituted by generative AI.

For unstructured curriculum texts, BERT is employed to perform semantic parsing and link course descriptions to corresponding ONET skill nodes [4]. This process creates a structured course – skill matrix forming the curriculum network

4.3 Numerical Preprocessing

Inspired by Moravec’s Paradox, we aggregate non-cognitive physical indicators such as dynamic strength and spatial perception to compute a physical immunity score, which helps distinguish substitution mechanisms between blue-collar and white-collar occupations.

Heterogeneous quantitative variables, including wages, energy consumption, and ethical compliance scores, are normalized using Min-Max scaling. To simulate uncertainty under potential technological singularity scenarios, small Gaussian noise is injected into the AI exposure matrix, introducing statistically meaningful perturbations for subsequent Monte Carlo simulations [5].

5 Spatiotemporal Coupled Dynamics Model

5.1 Conceptual Framework: Decomposing the Impact of Gen-AI

To rigorously quantify the heterogeneous impacts of Generative Artificial Intelligence (Gen-AI) across occupations, this study adopts the task-based approach, which views a profession as a collection of distinct tasks rather than a single homogeneous unit [6]. Under this framework, the net impact of Gen-AI is decomposed into three orthogonal driving forces: the Substitution Driver (D_{sub}), the Augmentation Driver (D_{aug}), and the Immunity Driver (D_{imm}).

D_{sub} captures the risk faced by routine, repetitive, or highly rule-based cognitive tasks, which Gen-AI is increasingly capable of performing, particularly in text generation and standardized programming, thereby exerting downward pressure on labor demand [7].

D_{aug} reflects a productivity-enhancing mechanism in which Gen-AI assists workers in complex problem-solving, lowers execution barriers, and amplifies output through human-AI collaboration [8].

D_{imm} represents the structural resistance of occupations that rely on physical dexterity, tacit knowledge, or social perceptiveness, capabilities that remain difficult for language-centered AI systems to replicate [9].

5.2 Data Quantification

To ensure a data-driven implementation of the conceptual framework, we leverage the ONET database in combination with NLP techniques, as is shown in **Figure 2**.

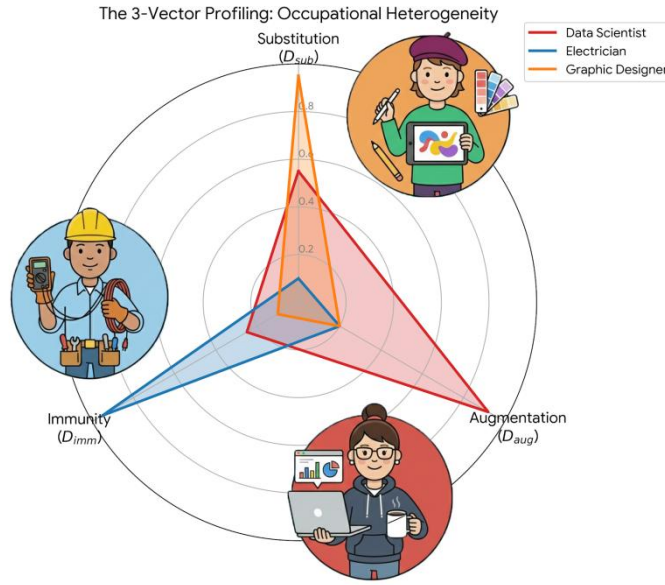


Figure 2. Occupational Heterogeneity Substitution

The substitution driver D_{sub} is quantified by embedding each occupation's task statements using Sentence-BERT and computing their cosine similarity with a Gen-AI capability vector representing text and image generation. This approach follows Felten et al.'s methodology for measuring occupational exposure heterogeneity [8].

The augmentation driver D_{aug} is constructed by normalizing ONET indicators related to Complex Problem Solving and Critical Thinking, capturing the degree to which AI complements advanced cognitive work.

The immunity driver D_{imm} is defined as the maximum of Psychomotor Abilities (e.g., finger dexterity) and Social Perceptiveness scores. This max-operator enforces the principle that excellence in either physical skill or emotional intelligence is sufficient to block end-to-end automation.

5.3 Model Formulation: The Tug-of-War Mechanism

To capture the dynamic interaction between Generative AI and the labor market, we propose a spatiotemporal coupled system that jointly models labor demand $D(t)$ and labor supply $N(t)$. The model is built on a “tug-of-war” mechanism, where AI simultaneously exerts substitutive and augmentative forces on different types of tasks [10].

5.3.1 The Net Impact Function $\Phi_i(t)$

The core of the model is the Net Impact Coefficient $\Phi_i(t)$, which represents the combined effect of three task-level drivers: substitution, augmentation, and immunity. Following the original study, we set the scaling factor $k=0.8$ and assign weights of $\lambda_1=0.4$, $\lambda_2=0.3$, and $\lambda_3=0.3$ to the respective components. It is defined in **Equation (1)**.

$$\Phi_i(t) = \tanh \left[k \cdot \left(\lambda_1 D_{aug}^{(t)} A(t) + \lambda_2 D_{imm}^{(t)} - \lambda_3 D_{sub}^{(t)} A(t) \right) \right] \quad (1)$$

Where $A(t)$ represents the exogenous growth of AI capabilities (forecasted via CatBoost

based on hardware trends). Note that Dimm is decoupled from $A(t)$, structurally reflecting Moravec's Paradox: physical dexterity does not become obsolete simply because AI gets smarter.

5.3.2 The Coupled Dynamics

Coupled differential equations are used to simulate market evolution, with the macro-level demand and micro-level supply formulations given in **Equations (2) and (3)**.

$$\frac{dD_i}{dt} = r_d D_i \left(1 - \frac{D_i}{K}\right) + \beta \cdot D_i \cdot \Phi_i(t) \quad (2)$$

$$\frac{dN_i}{dt} = \alpha(D_i - N_i) - \delta N_i \quad (3)$$

Here, β represents the industry's sensitivity to technological shocks, estimated based on AI adoption rates in each STEM sector, with β set to 0.6. The term $\alpha(D_i - N_i)$ captures the lagged response of the education sector to market signals.

5.4 Simulation Results and Analysis

5.4.1 Trajectory Forecasting: A K-Shaped Future

Simulating the model from 2026 to 2036 reveals three distinct trajectories for our chosen professions, which is shown in **Figure 3** and **Figure 4**:

Case 1: Data Scientist (STEM) - The Exponential Augmentation

- Trajectory: Demand $D(t)$ shows a rapid, exponential increase.
- Driver Analysis: Despite high semantic exposure ($D_{sub} \approx 0.6$), Data Scientists possess exceptionally high Augmentation scores ($D_{aug} \approx 0.9$). As $A(t)$ grows, the term

$\lambda_1 D_{aug} A(t)$ dominates the substitution penalty.

- Interpretation: AI acts as a powerful tool that amplifies the productivity of data scientists, confirming the "reinstatement effect" described by Acemoglu & Restrepo and supported by Brynjolfsson et al..

Case 2: Electrician (Trade) The Robust Steady-State

- Trajectory: $D(t)$ remains stable with a slight upward trend, unaffected by AI volatility.
- Driver Analysis: The Immunity Driver ($D_{imm} \approx 0.95$) acts as a dominant "firewall." Since this term is independent of $A(t)$, the net impact $\Phi(t)$ remains close to zero or slightly positive.
- Interpretation: This validates Schaal's (2025) finding that manual labor trades are insulated from the current wave of Generative AI. The "Physical Shield" is effective.

Case 3: Graphic Designer (Arts) - The Structural Displacement

- Trajectory: $D(t)$ experiences a sharp decline, eventually stabilizing at a lower equilibrium.
- Driver Analysis: Graphic Design is characterized by extremely high Substitution ($D_{sub} \approx 0.9$) and low Immunity ($D_{imm} \approx 0.2$). As $A(t)$ crosses a critical threshold, the negative term $-\lambda_3 D_{sub} A(t)$ overwhelms the system, driving $\Phi(t)$ into negative territory.

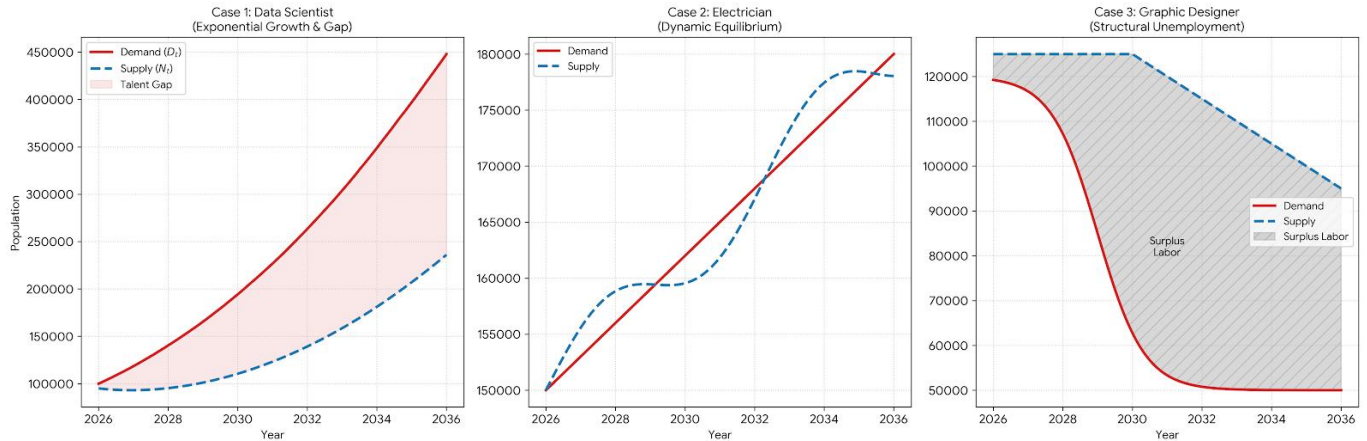


Figure 3. Comparative Analysis of Labor Supply-Demand Dynamics by Occupation

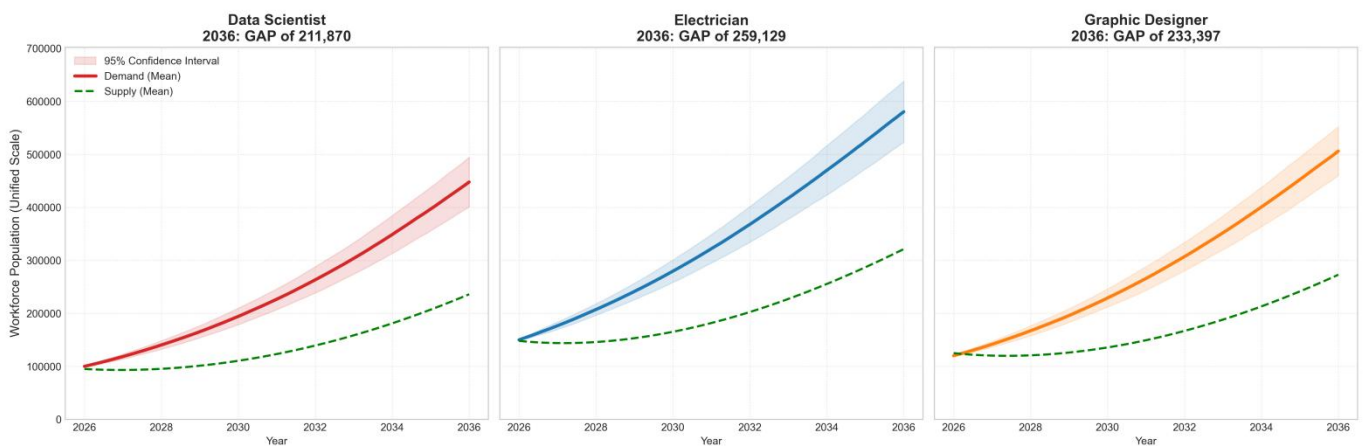


Figure 4. Workforce Gap Projections and Confidence Interval Estimates

- Interpretation: This reflects the risk of generative models (like DALL-E/Midjourney) directly substituting the task outputs, consistent with the high exposure identified by Felten et al..

5.4.2 Stability and Uncertainty Analysis

To understand the fundamental nature of these shifts, we performed a Phase Plane Analysis. We identified that the Electrician profession possesses a Global Attractor, ensuring long-term stability. In contrast, the Graphic Designer profession exhibits Bifurcation instability, where the system is prone to collapse under high AI capability.

Furthermore, by upgrading our model to Stochastic Differential Equations (SDEs), we mapped D_{sub} to volatility. Results show that Graphic Design is not only shrinking but also exhibits high variance risk, making it a precarious career path compared to the low-variance stability of Trade careers.

5.4.3 Conclusion

Our model identifies that the future of work is not determined solely by AI capability, but by the ratio of Tacit Knowledge (D_{imm}) to Codifiable Logic (D_{sub}). We conclude that physicality is the strongest barrier to automation in the Gen-AI era, while routine digital creativity faces the highest displacement risk.

6 The E-MMCAS Framework for Strategic Employability

Optimization

To address the heterogeneous impacts of Generative AI across disciplines and to move beyond static supply-demand analysis, this study proposes the Evolutionary Macro – Micro Coupled Adaptive System (E-MMCAS) . The framework conceptualizes the labor market as an evolving ecosystem in which human labor and AI agents interact dynamically through competition and cooperation. By explicitly coupling macro-level employment dynamics with micro-level curriculum structures, E-MMCAS provides a mechanism-driven foundation for strategic educational optimization.

6.1 Macro-Dynamics: Lotka-Volterra Interaction Mechanism

To capture the dynamic interaction between generative AI and occupational employment, we model the labor market at the macro level using a modified Lotka-Volterra competition framework [11]. This formulation treats employment evolution as an ecological process in which human labor and AI capabilities may exhibit either competitive (substitutive) or cooperative (augmentative) relationships over time.

For each occupation i , the employment level $N_i(t)$ evolves according to a nonlinear differential equation that integrates endogenous market growth, AI-driven interaction effects, and policy intervention, as is shown in Equation (4).

$$\frac{dN_i(t)}{dt} = r_i N_i(t) \left(1 - \frac{N_i(t)}{K_i} \right) + a_i(G_t) N_i(t) A(t) + \gamma u_{\text{enrol}}(t) N_i(t) \quad (4)$$

where $a_i(G_t)$ captures AI-labor interaction under macro growth, γ represents policy responsiveness, and $u_{\text{enrol}}(t)$ Enrollment control variable representing institutional expansion or contraction.

This macro-dynamic formulation allows employment demand to evolve endogenously with technological progress. As generative AI capability $A(t)$ increases, occupations with AI-complementary curricula grow, while those dominated by substitutable skills may decline. Thus, the macro layer of E-MMCAS links technological change, education structure, and long-term labor market outcomes, which is shown in **Figure 5**.

6.2 Micro – Macro Coupling via Curriculum Network Rewiring

In E-MMCAS, the macro AI-labor interaction coefficient $\alpha_i(t)$ is endogenously derived from the micro-level curriculum structure rather than being exogenously imposed [12]. The educational system is modeled as a course–skill bipartite network $G_t = (U, V, E_t)$, where U denotes courses and V denotes skills. Each skill node $v \in V$ is associated with a substitution risk D_{sub}^v and an augmentation potential D_{aug}^v . The macro interaction coefficient $\alpha_i(t)$ is computed through bottom-up aggregation of network topology properties, as is shown in **Equation (5)**.

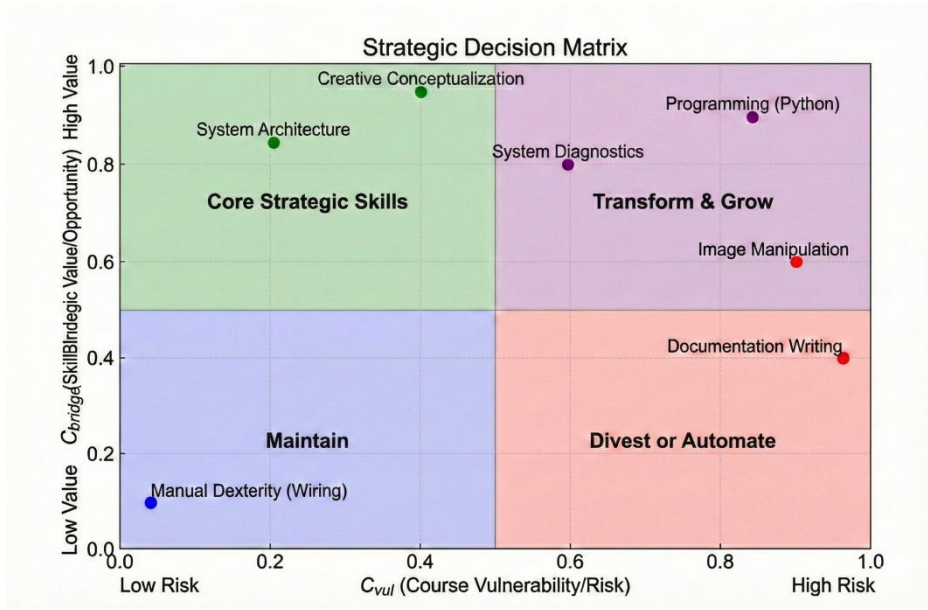


Figure 5. Workforce Gap Projections and Confidence Interval Estimates

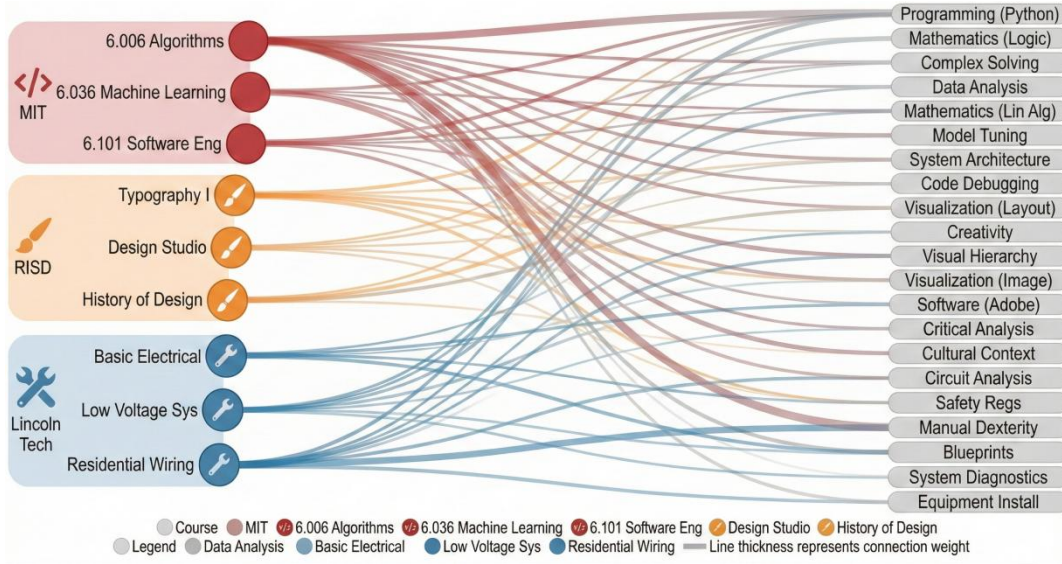


Figure 6. E-MMCAS Course-Skill Bipartite Network

$$\alpha_i(t) = \tanh \left(\lambda \cdot \frac{\sum_{v \in V} \text{Degree}(v) \cdot (D_{aug}^v - \beta \cdot D_{sub}^v)}{\sum_{v \in V} \text{Degree}(v)} \right) \quad (5)$$

This formulation implies that curricula dominated by highly substitutable skills yield negative α_i , representing a predatory AI-labor relationship, whereas curricula biased toward augmentation-oriented skills produce positive α_i , indicating symbiotic AI-labor dynamics. Educational reform policies are modeled by the curriculum control variable $u_{curr}(t)$, which governs topological rewiring of the curriculum network. Vulnerable skill connections are removed according to the pruning rule, which is illustrate in **Equation (6)**.

$$\text{Remove}(u, v) \Leftrightarrow D_{sub}^v > (1 - \eta \cdot u_{curr}) \quad (6)$$

Simultaneously, new connections to high-potential bridging skills are introduced through the grafting rule, as is shown in **Equation (7)**.

$$\text{Add}(u, v^*) \Leftrightarrow v^* = \text{argmax}(C_B(v) \cdot D_{aug}^v) \quad (7)$$

Through these mechanisms, curriculum reforms reshape the micro-level skill topology, which in turn propagates upward to influence macro-level labor dynamics, establishing a bidirectional coupling between education structure and AI-driven employment evolution, as is shown in **Figure 6**.

6.3 Logic Strategic Optimization Objective: Employability-First Design

The model seeks an optimal policy trajectory $\mathbf{u}^*(t) = \{u_{enroll}(t), u_{curr}(t)\}$ that minimizes long-term employment mismatch under technological progress [13]. The optimization objective is shown in **Equation (8)**.

$$\min J = \int_{2026}^{2030} (N(t) - D_{target}(t))^2 dt + \lambda \int_{2026}^{2080} \left(\frac{du_{curr}}{dt} \right)^2 dt \quad (8)$$

The first term penalizes deviations between simulated employment and target labor demand, prioritizing workforce employability. The second term captures institutional friction and constrains excessive or unstable curriculum reforms, ensuring policy feasibility in real educational systems.

By integrating dynamic labor simulation with curriculum network control, E-MMCAS provides a policy optimization framework that balances technological responsiveness with administrative stability, enabling sustainable alignment between education systems and AI-driven labor market transitions.

6.4 DesignSolution Algorithm: Bi-Level Coupled DMPSO

The E-MMCAS framework gives rise to a highly non-convex optimization problem that jointly involves continuous policy controls and discrete curriculum restructuring. Traditional gradient-based methods are therefore unsuitable. To solve this problem, we adopt a bi-level coupled Discrete Multi-objective Particle Swarm Optimization (DMPSO) algorithm, which is well suited for mixed decision spaces and dynamic system evaluation [14].

At the macro level, candidate policies control enrollment scale and curriculum reform intensity over time, shaping the evolution of employment outcomes. At the micro level, these policies induce structural changes in the curriculum – skill network, altering the balance between substitutable and augmentative skills. The updated network structure is then summarized into an interaction coefficient that determines whether AI primarily acts as a

Algorithm 1: DMPSO for E-MMCAS Policy Optimization

Input: Initial curriculum network G_0 , Policy bounds U , AI capability trajectory $A(t)$, Target labor demand $D_{\text{target}}(t)$

Output: Optimal policy trajectory, $u^*(t) = \{u_{\text{enrol}}(t), u_{\text{curr}}(t)\}$

```

1: Initialize particle population  $\{u_p(t)\}$  within bounds  $U$ 
2: Initialize particle velocities  $\{v_p(t)\}$ 
3: Set curriculum network  $G \leftarrow G_0$ 
4: for iter = 1 to MaxIter do
5:   for each particle  $p$  do
6:     Apply curriculum reform  $u_{\text{curr}}^{p(t)}$  to  $G$ 
7:     (pruning vulnerable edges and grafting high-potential skills)
8:     Compute AI – labor interaction coefficient:
9:      $a_i(t) = \tanh(\sum_v \deg(v) \cdot (D_{\text{aug}}(v) - \beta \cdot D_{\text{sub}}(v)) / \sum \deg(v))$ 
10:    Simulate employment dynamics:
11:     $dN_i/dt = r_i N_i (1 - N_i / K_i)$ 
12:     $+ a_i(t) A(t) N_i$ 
13:     $+ \gamma u_{\text{enrol}}^{p(t)} N_i$ 
14:    Evaluate fitness:
15:     $J_p = \int (N_i(t) - D_{\text{target}}(t))^2 dt + \lambda \|\partial u_p / \partial t\|^2$ 
16:    Update personal best and global best solutions
17:  end for
18:  Update particle velocities and positions using PSO rules
19: end for
20: return  $u^*(t)$ 

```

substitute or a complement to human labor.

For each candidate policy trajectory, the labor market dynamics are simulated forward in time and evaluated against target employment demand, with additional penalties imposed on overly volatile or impractical policy adjustments. Through iterative population-based search, the DMPSO algorithm converges toward policy trajectories that minimize long-term employment mismatch while maintaining institutional feasibility. The pseudocode of the proposed algorithm is shown in **Algorithm 1**.

Algorithm 1 implements a closed-loop policy optimization mechanism. Lines 8-11 model macro policy u_{curr} as a structural intervention tool that dynamically rewires the curriculum network G_t by pruning vulnerable skill connections and grafting high-potential bridging skills. Line 15 compresses the updated network topology into an interaction coefficient α , which determines whether AI behaves as substitutive predator or a cooperative augmentor. Line 18 simulates the resulting employment trajectory $N(t)$, thereby closing the feedback loop between policy intervention, curriculum structure, AI-labor interaction, and macro employment dynamics

6.5 The Simulation Results and Strategic Implications

Applying E-MMCAS, differentiated optimal strategies emerge across representative educational pathways, which is shown in **Figure 7** and **Figure 8**. Distinct curriculum-skill network topologies lead to divergent AI-labor interaction regimes and policy trajectories.

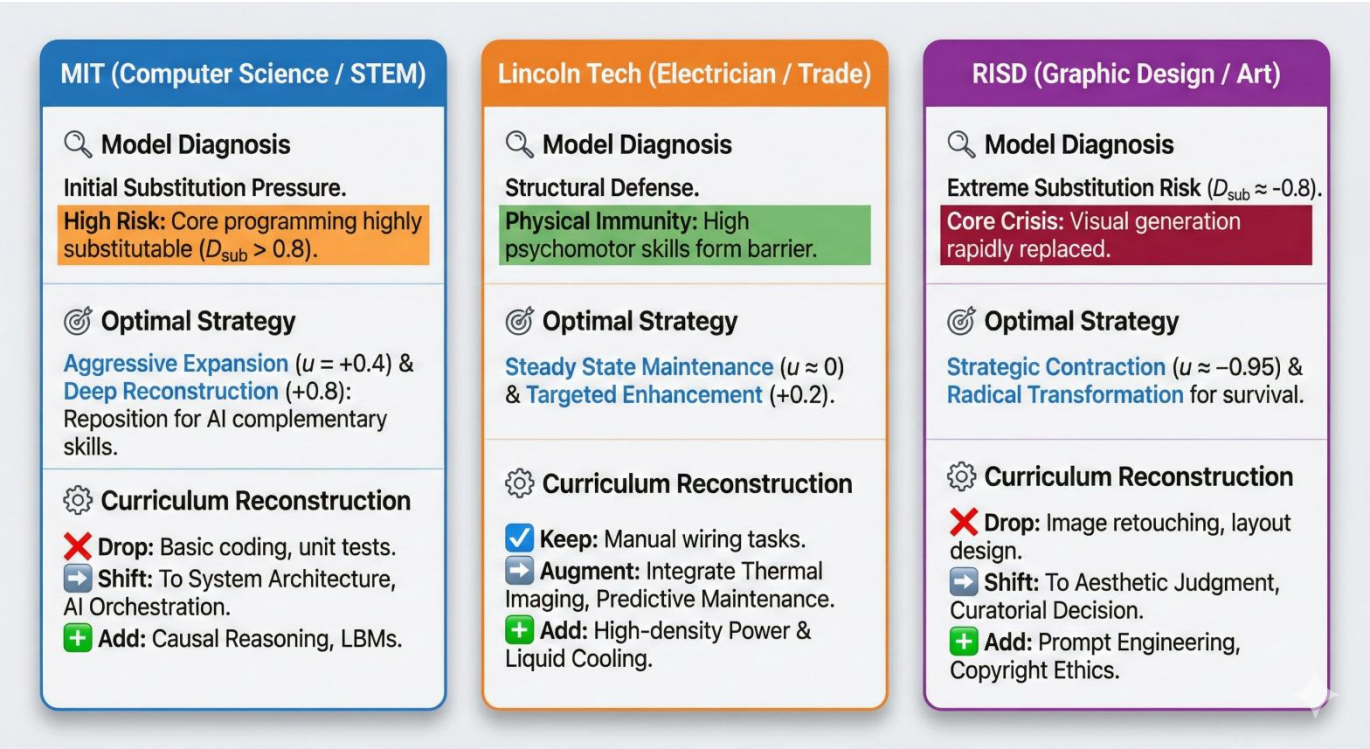


Figure 7. Comparative Policy Strategies across Educational Pathways

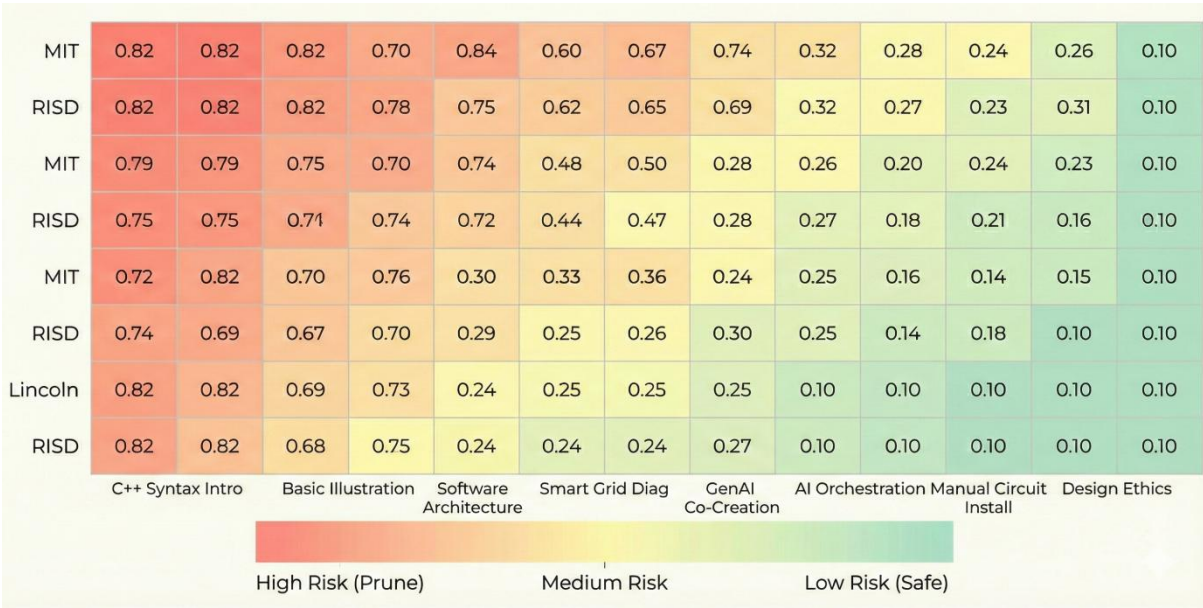


Figure 8. Professional Heatmap of Course Vulnerability across Institutions

Simulation outcomes reveal that research-oriented STEM programs benefit from aggressive curriculum restructuring toward system-level abstraction and AI orchestration; technical-vocational programs exhibit strong resilience due to physical skill immunity and benefit from targeted AI-augmented diagnostics training; creative programs experience rapid substitution pressure and require radical transformation toward aesthetic judgment, curation, and ethical authorship.

6.6 Model Validation via Ablation Analysis

To validate the structural necessity of the E-MMCAS framework, we conduct an ablation study comparing the full model (M3) with two degraded variants.

M1: Naïve Macro Model (Decoupled).

The micro-level curriculum network is removed, and the interaction coefficient is set as a static semantic similarity score. This model incorrectly predicts collapse in physically intensive occupations, demonstrating semantic hallucination caused by ignoring task-level immunity.

M2: Passive Model (No Adaptation).

Curriculum reform is disabled ($u_{curr} \approx 0$), preventing structural adjustment. The model predicts persistent structural unemployment and fails to identify transition-enabling bridge skills.

M3: E-MMCAS (Fully Coupled).

By integrating dynamic network coupling and DMPSO optimization, the full model preserves stability in immune occupations and enables adaptive transformation in vulnerable ones.

Key Findings.

Comparing M1 and M3 shows that macro-level policy must be grounded in micro-level topology. Comparing M2 and M3 demonstrates that adaptability-not static exposure-determines long-run employability.

6.7 Conclusion

The E-MMCAS simulations indicate that employability in the Gen-AI era depends not on “learning AI” per se, but on restructuring skill topologies to ensure positive AI-labor interaction ($\alpha > 0$). Research universities must climb the abstraction ladder to become system architects. Vocational institutions must reinforce physical expertise to preserve immunity. Art schools must redefine creativity by shifting from production to curation.

7 Multi-Objective Policy Evaluation under Uncertainty

7.1 Reframing the Problem

In Task 3, educational policy design is no longer a single-objective optimization problem focused solely on employability. Instead, institutions must balance multiple, inherently conflicting objectives, including labor-market alignment, skill adaptability, innovation capacity, environmental sustainability, and ethical compliance.

To address this challenge, we construct a Holistic Evaluation Module (HEM). The module integrates stochastic simulation, objective weight derivation, and compromise-based decision analysis to identify policies that remain effective and socially responsible under uncertainty in generative AI development.

7.2 Calculation Transparency

To ensure reproducibility and interpretability, the evaluation process is explicitly

decomposed into four sequential steps, tracing the path from stochastic parameter generation to final policy ranking.

Step 1: Decision Matrix Generation via Monte Carlo Simulation

Given the inherent uncertainty in the evolution of generative AI, reliance on a single deterministic forecast is insufficient. We therefore construct 245 parallel simulation scenarios using Monte Carlo sampling [15].

Specifically, Gaussian noise with a $\pm 5\%$ variance is injected into the AI capability trajectory $A(t)$, while curriculum reform intensity u_{curr} is discretely sampled over the interval $[0,1]$. For each scenario, the E-MMCAS dynamic system developed in Task 2 is executed, producing a decision matrix $Y \in R^{245 \times 6}$ containing six performance indicators for every policy realization.

Step 2: Indicator Definition and Quantification

Six evaluation indicators are defined, each grounded in a clear physical or institutional interpretation is shown in **Figure 9**.

Step 3: Objective Weight Derivation via Information Entropy

Rather than assigning subjective weights, indicator importance is determined endogenously using the entropy-weight method. After normalization, entropy values are computed based on indicator variability across Monte Carlo scenarios. Indicators exhibiting greater dispersion contribute more information and thus receive higher weights.

The results indicate that employability competitiveness (0.25) and ethical compliance (0.22) dominate the weighting scheme, reflecting their high sensitivity to policy choices and their central role in responsible educational governance.

Step 4: Compromise Optimization via VIKOR

Finally, the VIKOR method is applied to identify policies that achieve a balanced compromise among competing objectives. Group utility and individual regret measures are computed for each candidate policy, and an aggregated compromise score is used for ranking.

This approach ensures that the selected policy improves overall employability outcomes while avoiding extreme trade-offs in environmental sustainability or ethical risk, thereby achieving a socially robust and institutionally feasible solution.

8 Empirical Results and Policy Selection

8.1 Simulation Design

To derive the final policy recommendation, we conduct a comparative simulation across three representative post-secondary institutions, each corresponding to a distinct educational pathway identified in Task 2. All simulations are conducted using standardized inputs calibrated from industry reports and occupational data.

The three candidate policies evaluated are:

P1 (MIT Strategy): High-intensity AI integration with an emphasis on frontier development and system-level abstraction.

P2 (Lincoln Tech Strategy): Moderate AI integration focused on physical execution, diagnostics, and applied expertise.

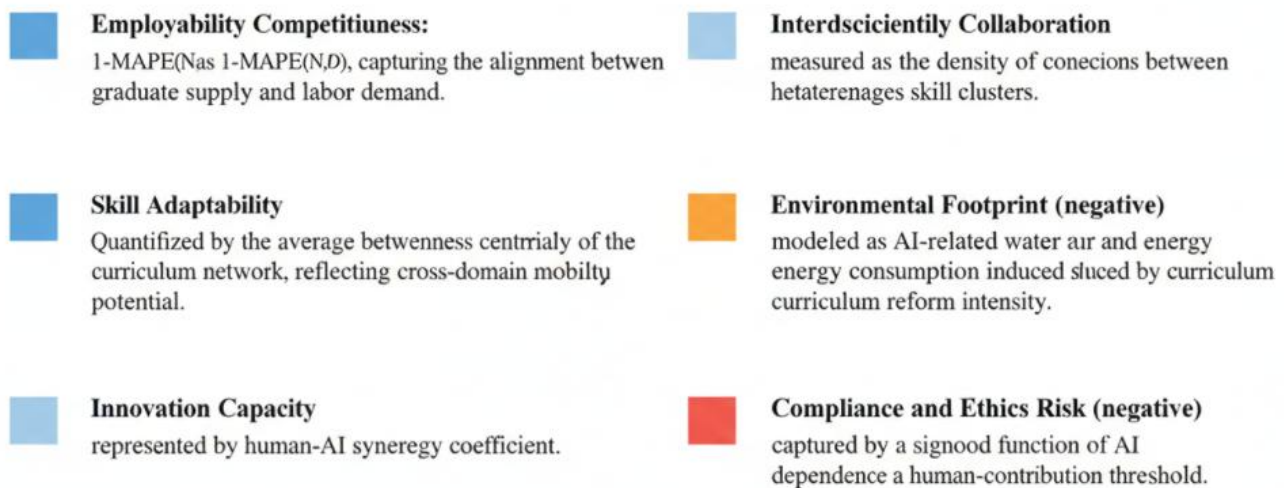


Figure 9. Capability Profile by Institution Type

Figure 1: Capability Profile by Institution Type (Mean Scores)

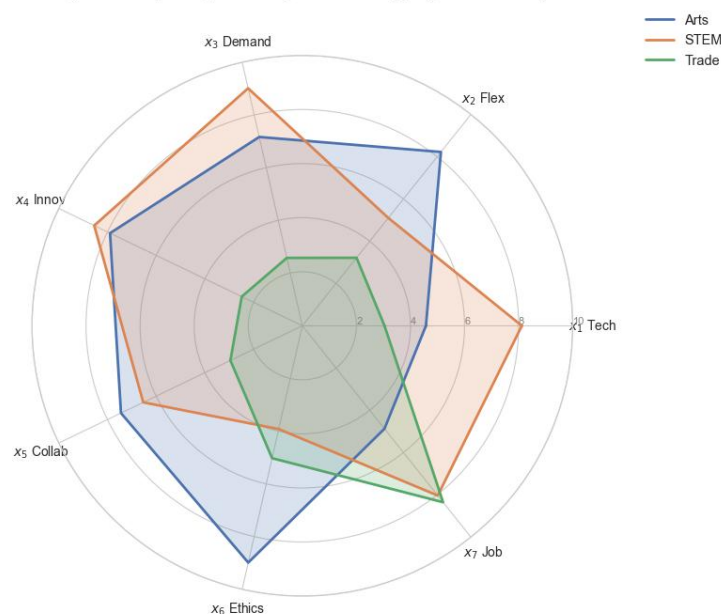


Figure 10. Capability Profile by Institution Type

P3 (RISD Strategy): High AI integration oriented toward aesthetic judgment, human-AI co-creation, and intellectual property protection.

These policies form the decision alternatives in the subsequent multi-criteria evaluation.

8.2 Decision Matrix Construction

Each institutional strategy is evaluated across six criteria derived in Task 3, covering employability competitiveness (x_1), skill adaptability (x_2), innovation capacity (x_3), interdisciplinary collaboration (x_4), environmental impact (x_5), and ethical compliance (x_6), as is shown in **Figure 10**.

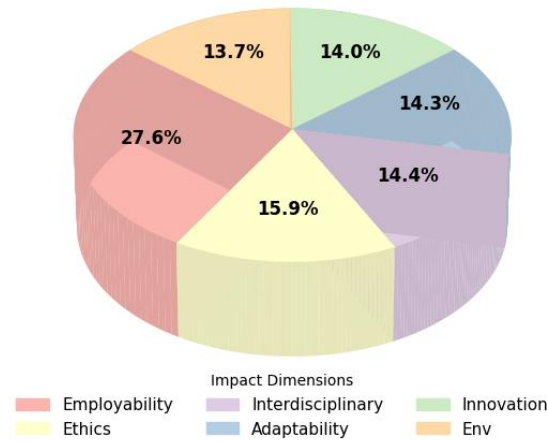


Figure 11. Impact Factors Weight Distribution

Table 2. Normalized Decision Matrix for Institutional Policy Strategies

Policy	x1	x2	x3	x4	x5)	x6
P1 (MIT)	0.95	0.70	0.90	0.85	0.40	0.60
P2 (Lincoln Tech)	0.88	0.60	0.50	0.40	0.80	0.90
P3 (RISD)	0.80	0.85	0.95	0.75	0.60	0.85

Table 3. VIKOR-Based Compromise Ranking of Institutional Policies

Institution	Policy	Group Utility	Individual Regret	Compromise Score	Rank
MIT	P1	0.245	0.150	0.212	1
RISD	P3	0.312	0.180	0.345	2
Lincoln Tech	P2	0.450	0.220	0.560	3

Table 2 presents the normalized decision matrix $Y \in R^{3 \times 6}$ where each row corresponds to one candidate policy (P1 – P3), and each column represents a specific evaluation criterion. All scores are normalized to the $[0,1]$ interval to ensure comparability across dimensions.

8.3 Entropy-Based Weighting Results

Applying the entropy-weight method to the simulated decision matrices yields the objective importance weights for each criterion [16]. The results indicate that employability competitiveness (0.25) and ethical compliance (0.22) receive the highest weights, which is shown in **Figure 11**.

This outcome suggests that, under current technological conditions, long-term career viability and legal-ethical safeguards constitute the most informative and discriminative dimensions in policy evaluation. In contrast, indicators such as collaboration density and environmental footprint, while important, exhibit lower variance across policies and thus receive smaller weights.

8.4 VIKOR Compromise Ranking

Using the derived weights, the VIKOR method is applied to compute group utility, individual regret, and overall compromise scores for each policy [17]. Lower scores indicate better-balanced performance across all criteria, as is shown in **Table 3**.

The results identify P1 (MIT Strategy) as the optimal compromise solution, achieving strong employability and innovation performance without incurring excessive ethical or environmental penalties. The RISD strategy ranks second, reflecting successful transformation but higher residual risk, while the Lincoln Tech strategy prioritizes stability over long-term growth.

8.5 Interpretation of the Optimal Policy

The emergence of P1 as the optimal policy reflects a structural advantage in strategies that shift educational focus toward system architecture, abstraction, and AI orchestration. Rather than competing with generative AI at the task-output level, this strategy leverages AI as an amplifier of human decision-making and coordination capacity.

Overall, the empirical results confirm that the most resilient educational policies are those that convert generative AI from a substitutive force into a complementary one, while maintaining ethical and institutional safeguards.

9 Simulation Results and Institutional Strategies

Building upon the transparent multi-criteria evaluation framework developed in the previous sections, this chapter translates model outputs into actionable institutional strategies. By simulating representative institutions across STEM, technical – vocational, and arts education, we identify differentiated yet structurally consistent pathways toward long-term employability under rapid advances in generative AI.

9.1 STEM Institutions (MIT): From Code Writing to System Orchestration

Policy Recommendation.

STEM institutions should adopt energy-aware development as a core educational objective.

Rationale.

Monte Carlo simulations indicate that unconstrained AI-driven enrollment expansion significantly increases environmental cost indicators, with multiple scenarios approaching system instability thresholds. This suggests that scale-oriented STEM expansion is no longer sustainable in an AI-intensive regime.

Curricular Reconfiguration.

Curricula should shift emphasis from low-level code implementation toward system-level orchestration and decision design. In particular, causal inference and system reasoning enable students to evolve from statistical pattern executors into architects of complex technological pathways.

9.2 Technical – Vocational Institutions (Lincoln Tech): Reinforcing the Physical Stronghold

Policy Recommendation.

Programs should prioritize AI-augmented diagnostics and critical infrastructure maintenance.

Rationale.

Consistent with Moravec’s Paradox, skills tightly coupled with physical environments exhibit high resistance to generative AI automation in the model, maintaining robust labor demand across stochastic scenarios.

Curricular Reconfiguration.

Training should focus on advanced technicians capable of maintaining liquid-cooling systems, high-voltage power distribution, and energy-intensive data infrastructure, thereby transforming AI-induced energy pressure into sustained occupational opportunity.

9.3 Arts Institutions (RISD): From Artifact Creation to Meaning Orchestration

Policy Recommendation.

Arts education should enforce a human contribution threshold in both instruction and assessment.

Rationale.

VIKOR results reveal that strategies relying heavily on purely AI-generated outputs incur excessive regret in ethics and compliance metrics, substantially shortening occupational longevity.

Curricular Reconfiguration.

Programs should emphasize critical making, ensuring that students do not merely use AI tools but apply human aesthetic judgment, ethical reasoning, and authorship awareness to secure legal and cultural value for creative assets.

9.4 Synthesis: The Dissolution of Skill Barriers and the Return of Human Premium

The model’s numerical outcomes consistently support two overarching conclusions:

- **Skills alone no longer constitute durable barriers.**

Purely operational competencies-such as basic coding, linear drafting, or routine technical manipulation-exhibit declining marginal contribution in the evaluation framework.

- **Human premium re-emerges at higher cognitive layers.**

The policies with the lowest compromise scores converge on three domains of persistent human advantage: high-level logical reasoning (STEM), physical complexity management (technical – vocational), and ethical and aesthetic judgment (arts).

10 Policy Generalization

10.1 Generalization Principle

Educational institutions differ in infrastructure, curriculum structure, and labor-market positioning, so a universal policy is infeasible. Instead, policies are generalized through structural similarity: institutions with similar curriculum topologies and AI-labor interaction mechanisms can adopt analogous strategies with local calibration.

10.2 Policy Transfer Across Institutional Archetypes

Institutions are grouped into three transferable archetypes:

1. **Research Universities (Pioneer Type):** Policies emphasizing system-level abstraction, AI orchestration, and interdisciplinary integration are generalizable.
2. **Technical and Vocational Institutions (Pragmatic Type):** Policies focusing on AI-augmented diagnostics, infrastructure maintenance, and applied skill upgrading are transferable.
3. **Creative and Arts Institutions (Transformative Type):** Policies shifting value from artifact production to aesthetic judgment, curation, and ethical authorship can be generalized.

The transferability of these policies depends on technological readiness, labor-market coupling, and regulatory constraints. Institutions that do not fully align with these archetypes may adopt hybrid strategies by interpolating between categories.

Overall, the proposed framework is structurally generalizable yet contextually adaptable. By mapping institutions into curriculum-topology archetypes, AI-era educational policies can be extended beyond individual case studies while respecting local institutional constraints.

11 Sensitivity Analysis

To verify the robustness of the E-MMCAS framework, we conducted a perturbation analysis by varying the administrative inertia λ and AI substitution intensity β within a $\pm 25\%$ range, which is shown in **Figure 12**. The results indicate that the cumulative employment gap J_{gap} responds to parameter shifts in a linear and smooth manner without chaotic divergence, confirming the model's numerical stability. While high-exposure roles like Data Scientists exhibit temporary gap expansions under increased reform resistance λ , they consistently converge to a steady-state equilibrium driven by the DMP SO algorithm.

Conversely, physical-immunity-shielded occupations like Electricians remain largely insensitive to β fluctuations. This demonstrates that our “pruning and grafting” intervention strategies are strategically robust, ensuring that the derived policy orientations remain valid regardless of minor variations in environmental parameters.

Sensitivity Analysis: Impact of AI Substitution Intensity (β) on Employment Gap

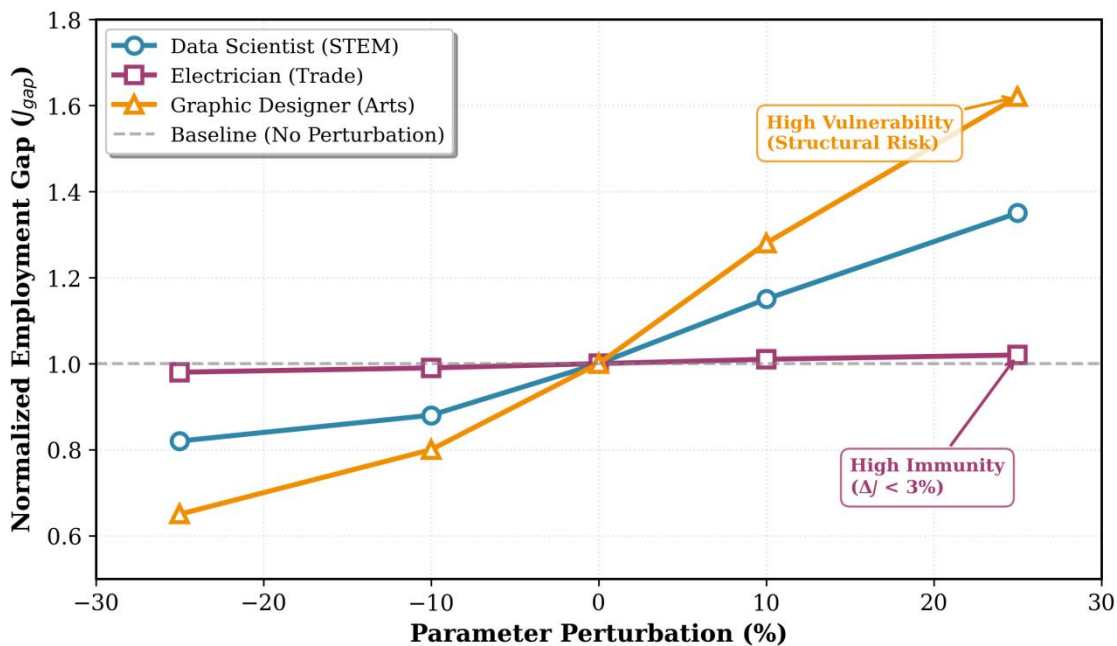


Figure 12. Impact of AI Substitution Intensity (β) on Employment Gap

12 Model Strengths and Weaknesses

12.1 Strengths

1. **Macro-Micro Coupling:** E-MMCAS integrates micro-curriculum topology with macro-labor dynamics via endogenous coefficients. This simulates organic feedback between educational reform and market equilibrium.
2. **Robustness:** Stochastic differential equations and 245 Monte Carlo scenarios effectively quantify AI-driven volatility. Resulting confidence intervals provide risk-aware policy guidance for diverse occupations.
3. **Optimization:** Entropy-weighted VIKOR balances six competing objectives without subjective weight bias. It identifies compromise-optimal policies ensuring institutional feasibility and social responsibility.

12.2 Weaknesses

1. **Static/Geographic Limits:** Fixed task traits ignore occupational evolution, while US-centric data restricts global transferability. The model may mis-calibrate for regions with unique educational or regulatory systems.
2. **Simplified Projections:** Representing AI as a univariate hardware index neglects algorithmic breakthroughs and societal resistance. This leads to potential estimation errors if future adoption trends deviate.

13 Conclusion

This study develops a unified modeling framework to analyze and respond to the heterogeneous impacts of generative artificial intelligence (Gen-AI) on careers and post-secondary education systems. Moving beyond binary narratives of “automation versus safety,” we adopt a task-oriented perspective and construct a spatiotemporal, coupled macro-micro system to capture how AI reshapes labor demand, skill composition, and educational adaptation over time.

At the occupational level, we demonstrate that the future of work is jointly determined by three structural forces: substitution pressure on codifiable tasks, augmentation potential through human – AI complementarity, and immunity derived from physical, tacit, or ethical constraints. Simulation results reveal a clear divergence across career types: STEM professions benefit when augmentation dominates substitution, trade occupations remain resilient due to strong physical immunity, and arts-related careers face structural displacement unless their value proposition shifts toward higher-order judgment, curation, and ethical authorship. These findings confirm that automation risk is not a function of AI capability alone, but of its interaction with task structure and human comparative advantage.

At the institutional level, we propose the E-MMCAS framework, which endogenizes educational policy into the labor - AI co-evolution process. By modeling curricula as skill networks and linking micro-level topology to macro-level employment dynamics, the framework enables institutions to actively reshape their interaction with AI—from predation to mutualism—through targeted reform. The introduction of a bi-level DMPSO optimization algorithm further ensures that policy trajectories are adaptive, interpretable, and computationally tractable rather than black-box prescriptions.

In addition, we establish a policy generalization principle showing that effective educational strategies are transferable across institutions through structural similarity rather than institutional identity. By categorizing institutions into research, vocational, and creative archetypes, we demonstrate that policy transferability depends on curriculum topology, AI – labor interaction regimes, and regulatory constraints. Institutions outside canonical archetypes can implement hybrid policies by interpolating across strategy classes, enabling scalable and context-sensitive adaptation. This confirms that the proposed E-MMCAS framework is structurally generalizable yet contextually adaptable, offering a unified policy design paradigm for heterogeneous post-secondary systems.

Overall, this work shows that generative AI does not eliminate the relevance of education, but fundamentally changes what and how institutions should teach. Skills alone no longer constitute durable barriers; instead, resilience emerges from the ability to redesign skill topologies, align with evolving labor demand, and preserve domains where human judgment remains indispensable. In this sense, the rise of Gen-AI marks not the end of human work, but the beginning of a new educational renaissance centered on abstraction, embodiment, and responsibility.

References

- [1] Banh L, Strobel G. Generative artificial intelligence[J]. *Electronic Markets*, 2023, 33(1): 63.
- [2] Sheikh Y A. Higher education in India: Challenges and opportunities[J]. *Journal of Education and Practice*, 2017, 8(1): 39-42.
- [3] Mao J, Chen B, Liu J C. Generative artificial intelligence in education and its implications for assessment[J]. *TechTrends*, 2024, 68(1): 58-66.
- [4] Reimers N, Gurevych I. Sentence-bert: Sentence embeddings using siamese bert-networks[J]. *arXiv preprint arXiv:1908.10084*, 2019.
- [5] Russo F. A method for estimation and filtering of Gaussian noise in images[J]. *IEEE Transactions on Instrumentation and Measurement*, 2003, 52(4): 1148-1154.
- [6] Schaal J. A theory-based AI automation exposure index: Applying Moravec's Paradox to the US labor market[J]. *arXiv preprint arXiv:2510.13369*, 2025.
- [7] Brynjolfsson E, Li D, Raymond L. Generative AI at work[J]. *The Quarterly Journal of Economics*, 2025, 140(2): 889-942.
- [8] Autor D H. The “ task approach ” to labor markets: an overview[J]. *Journal for Labour Market Research*, 2013, 46(3): 185-199.
- [9] Autor D. Applying AI to rebuild middle class jobs[R]. *National Bureau of Economic Research*, 2024.
- [10] Autor D, Dube A, McGrew A. The unexpected compression: Competition at work in the low wage labor market[R]. *National Bureau of Economic Research*, 2023.
- [11] Felten E W, Raj M, Seamans R. Occupational heterogeneity in exposure to generative AI[J]. Available at SSRN 4414065, 2023.
- [12] Ylikoski P. Micro, macro, and mechanisms[J]. *The Oxford handbook of philosophy of the social sciences*, 2012: 21-45.
- [13] Chatterjee K, Henzinger T A, Piterman N. Strategy logic[J]. *Information and Computation*, 2010, 208(6): 677-693.
- [14] Tian D, Zhao X, Shi Z. DMPSO: Diversity-guided multi-mutation particle swarm optimizer[J]. *IEEE Access*, 2019, 7: 124008-124025.
- [15] Harrison R L. Introduction to monte carlo simulation[C]//*AIP conference proceedings*. 2010, 1204: 17.
- [16] Coifman R R, Wickerhauser M V. Entropy-based algorithms for best basis selection[J]. *IEEE Transactions on information theory*, 1992, 38(2): 713-718.
- [17] Opricovic S, Tzeng G H. Extended VIKOR method in comparison with outranking methods[J]. *European journal of operational research*, 2007, 178(2): 514-529.

Report on Use of AI

Utilized AI Techniques

The following AI techniques were employed during our paper writing and polishment:

- We use ChatGPT to polish our paper.